

Fidelity witness notes

Companion papers: Gluza et al., [arXiv:1703.03152](https://arxiv.org/abs/1703.03152), Eisert et al., [arXiv:1910.06343](https://arxiv.org/abs/1910.06343)

0. Motivation: how do we verify many good qubits?

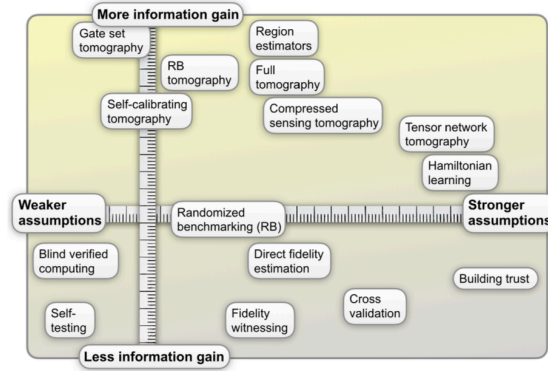
A quantum simulator claims to have prepared some many-body state. **Did it?**

Three challenges:

- **Scalability** — we want many qubits, but full tomography costs $\Theta(4^L)$ parameters.
- **Precision** — many pulses / deep circuits accumulate error.
- **Conceptual** — we want to *efficiently certify the quality of state preparation*, not learn the full state.

The Eisert–Hangleiter–Kashefi chart organizes verification methods along two axes: **information gain** vs **strength of assumptions**. Full tomography sits at maximal information / moderate assumptions; randomized benchmarking gives less information cheaply; **fidelity witnessing sits deliberately at low information gain, weak assumptions** — it answers a single question (a lower bound on fidelity) but does so scalably and with essentially no assumptions on the noise.

The core trade of this whole subject: give up learning the state; keep only a *certified scalar*.



1. Fidelity of state preparation

1.1 Definition and the two regimes

Subscripts throughout: **t** = **target** (the state you designed — pure by construction), **p** = **prepared** (what the machine actually made — mixed by physics).

$$F(\varrho_t, \varrho_p) = \begin{cases} \left(\text{tr} \left[(\sqrt{\varrho_t} \varrho_p \sqrt{\varrho_t})^{1/2} \right] \right)^2 & \text{in general} \\ \text{tr}[\varrho_t \varrho_p] & \varrho_t \text{ pure} \end{cases}$$

1.2 Making sense of it

(i) **Both pure.** $\varrho_t = |\psi_t\rangle\langle\psi_t|$, $\varrho_p = |\psi_p\rangle\langle\psi_p|$. Pure states are projectors, so $\sqrt{\varrho_t} = \varrho_t$, and the sandwich collapses:

$$\sqrt{\varrho_t} \varrho_p \sqrt{\varrho_t} = |\psi_t\rangle \underbrace{|\langle\psi_t|\psi_p\rangle|^2}_{\text{scalar}} \langle\psi_t| \implies F = |\langle\psi_t|\psi_p\rangle|^2$$

The familiar overlap. The modulus-squared kills the unphysical global phase.

(ii) **Target pure, preparation mixed. (this is what we would use)** Write $\varrho_p = \sum_k p_k |\phi_k\rangle\langle\phi_k|$ and average the pure-state overlaps:

$$F = \sum_k p_k |\langle\psi_t|\phi_k\rangle|^2 = \langle\psi_t|\left(\sum_k p_k |\phi_k\rangle\langle\phi_k|\right)|\psi_t\rangle = \langle\psi_t|\varrho_p|\psi_t\rangle = \text{tr}[\varrho_t\varrho_p]$$

("fidelity is the sum of diagonals of $\varrho_t\varrho_p$).

Note:

1. F = probability that the projective test $\{|\psi_t\rangle\langle\psi_t|, 1 - |\psi_t\rangle\langle\psi_t|\}$ answers *yes* on a single shot.
2. F = expectation value of the observable ϱ_t in the state ϱ_p .
3. F is **linear in ϱ_p** . This is the structural fact the entire witness formalism is built on.

1.3 Why not just measure F directly?

Since $F = \langle\varrho_t\rangle_{\varrho_p}$ is an expectation value, why not measure it? **Cost.** Decompose in the Pauli basis $P \in \{I, X, Y, Z\}^{\otimes n}$:

$$\varrho_t = \frac{1}{2^n} \sum_P c_P P, \quad c_P = \text{tr}[\varrho_t P] \implies F = \frac{1}{2^n} \sum_P c_P \langle P \rangle_{\varrho_p}$$

Up to 4^n terms. Using $\text{tr}[P_1 P_2] = 2^n \delta_{P_1 P_2}$ and purity $\text{tr}[\varrho_t^2] = 1$:

$$\boxed{\sum_P c_P^2 = 2^n} \quad (\text{purity sum rule})$$

A **stabilizer state** concentrates this budget in 2^n terms with $c_P = \pm 1$ (sparse). A **generic pure state** spreads it over all 4^n terms, each of typical size $|c_P| \sim 2^{-n/2}$ — exponentially many, individually tiny coefficients.

Sample count bottleneck. Estimate each $\langle P \rangle$ with N_P shots; since P has eigenvalues ± 1 , $\text{Var}(\hat{\mu}_P) \leq 1/N_P$. Independent estimates add in variance:

$$\text{Var}(\hat{F}) \leq \frac{1}{4^n} \sum_P \frac{c_P^2}{N_P}$$

Minimize over the allocation $\{N_P\}$ at fixed total $N = \sum_P N_P$ (Cauchy-Schwarz / Lagrange): the optimum is $N_P \propto |c_P|$, giving

$$\text{Var}(\hat{F}) = \frac{(\sum_P |c_P|)^2}{4^n N}$$

So everything hinges on the ℓ_1 **norm** $\sum_P |c_P|$. For a generic state, Cauchy-Schwarz gives $\sum_P |c_P| \leq \sqrt{4^n \cdot 2^n} = 2^{3n/2}$, and generic states saturate it:

$$\text{Var}(\hat{F}) \sim \frac{2^n}{N} \implies \boxed{N_{\text{total}} = O(2^n / \epsilon^2)}$$

Exponential. For a stabilizer target the same formula gives $N = O(1/\epsilon^2)$. The witness will beat this for *dense* targets by sampling in a different basis (mode space) where the target is sparse again.

2. Fidelity witness: definition and structure

2.1 Definition

An observable $\mathcal{W}$ is a **fidelity witness** for a *pure* state ϱ_t if

- (i) $\text{tr}[\mathcal{W}\varrho_p] = 1$ iff $\varrho_p = \varrho_t$
- (ii) $\text{tr}[\mathcal{W}\varrho_p] \leq \text{tr}[\varrho_t\varrho_p] \equiv F$ for all states ϱ_p

What we want in practice: measure $\langle \mathcal{W} \rangle_{\varrho_p} = 0.99$ and conclude $F \geq 0.99$.

The witness is a certified *lower bound* on the true fidelity — by definition, irrespective of what state the machine produced.

2.2 The operator inequality $\mathcal{W} \preceq \varrho_t$

Rearrange (ii): $\text{tr}[(\mathcal{W} - \varrho_t)\varrho_p] \leq 0$ for all ϱ_p — in particular for every pure $\varrho_p = |\phi\rangle\langle\phi|$:

$$\langle \phi | (\mathcal{W} - \varrho_t) | \phi \rangle \leq 0 \quad \forall |\phi\rangle \quad \Longleftrightarrow \quad \boxed{\mathcal{W} \preceq \varrho_t}$$

(Löwner order: $A \preceq B$ means $B - A$ is positive semidefinite (PSD), i.e. $\langle \phi | A | \phi \rangle \leq \langle \phi | B | \phi \rangle$ in *every* direction.

Consequences:

- $\mathcal{W}$'s expectation value never exceeds ϱ_t 's, in any state.
- (Weyl monotonicity) sorting both spectra descending, $\lambda_k(\mathcal{W}) \leq \lambda_k(\varrho_t)$ for every k .

2.3 The gap

Define $A := \varrho_t - \mathcal{W} \succeq 0$. Then for any preparation,

$$\boxed{F - \langle \mathcal{W} \rangle_{\varrho_p} = \text{tr}[A\varrho_p] \geq 0} \quad (\text{the systematic gap})$$

Interpretation: the witness under-reports by exactly the overlap of your noisy state with A . A big gap can push a genuinely good state below a certification threshold — a **false rejection**. It can never cause a false accept.

Tightening the bound = shrinking A = making $\mathcal{W}$ closer to ϱ_t = more expensive to measure.

2.4 Spectral structure of any witness

Since $\varrho_t = |\psi_t\rangle\langle\psi_t|$ is pure, applying it to $|\phi\rangle = c|\psi_t\rangle + |\phi_\perp\rangle$ shows it is a **rank-1 projector** with spectrum $\{1, 0, \dots, 0\}$.

(note: $\text{rank}(A) = \dim(\text{Im}(A))$)

Two properties follow for any valid $\mathcal{W}$:

Property 1 — $|\psi_t\rangle$ is an eigenvector of $\mathcal{W}$ with eigenvalue 1.

Proof. With $A = \varrho_t - \mathcal{W}$, condition (i) gives $\langle \psi_t | A | \psi_t \rangle = 1 - 1 = 0$. Now use the

🔴 **PSD Lemma.** If $A \succeq 0$ and $\langle \psi | A | \psi \rangle = 0$, then $A|\psi\rangle = 0$

Proof: $A = B^\dagger B$; then $\|B|\psi\rangle\|^2 = \langle \psi | A | \psi \rangle = 0$, so $B|\psi\rangle = 0$, so $A|\psi\rangle = 0$.

Hence $A|\psi_t\rangle = 0$, i.e. $\mathcal{W}|\psi_t\rangle = \varrho_t|\psi_t\rangle = |\psi_t\rangle$. ■

Property 2 — $\mathcal{W}$ is negative semidefinite on the orthogonal complement.

Proof. For $|\phi_\perp\rangle \perp |\psi_t\rangle$: $\langle \phi_\perp | \mathcal{W} | \phi_\perp \rangle \leq \langle \phi_\perp | \varrho_t | \phi_\perp \rangle = 0$. And $\mathcal{W}$ maps the complement into itself: since $\mathcal{W}$ is Hermitian and $\mathcal{W}|\psi_t\rangle = |\psi_t\rangle$,

$$\langle \psi_t | \mathcal{W} | \phi_\perp \rangle = (\mathcal{W}|\psi_t\rangle)^\dagger |\phi_\perp\rangle = \langle \psi_t | \phi_\perp \rangle = 0$$

Thus, no leakage out means no leakage in. So $\mathcal{W}$ block-diagonalizes and the complement block B has all eigenvalues ≤ 0 .

■

$$\mathcal{W} = 1 \cdot |\psi_t\rangle\langle\psi_t| + \sum_{k \geq 1} w_k |e_k\rangle\langle e_k|, \quad w_k \leq 0, \quad |e_k\rangle \perp |\psi_t\rangle$$

$$\mathcal{W} \cong \begin{pmatrix} 1 & 0 \cdots 0 \\ 0 & \\ \vdots & B \ (\preceq 0) \\ 0 & \end{pmatrix} \quad (\text{basis } |\psi_t\rangle, |e_1\rangle, \dots)$$

Thus, a fidelity witness is any Hermitian operator with eigenvalue +1 on the target and non-positive eigenvalues on everything orthogonal to it. The freedom in how negative the other eigenvalues go is what buys measurement cheapness and costs gap.

3. Worked example: Bell state

Target $\varrho_t = |\Phi^+\rangle\langle\Phi^+|$, $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Candidate:

$$\mathcal{W} = \frac{1}{2}(XX + ZZ)$$

Check via $A = \varrho_t - \mathcal{W}$. Expanding in the computational basis (or just diagonalizing):

$$\varrho_t - \mathcal{W} = |\Psi^-\rangle\langle\Psi^-|, \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

A projector, eigenvalues $\{1, 0, 0, 0\} \succeq 0 \implies \mathcal{W} \preceq \varrho_t \checkmark$. And $\mathcal{W}|\Phi^+\rangle = |\Phi^+\rangle \checkmark$.

The witness is not unique, and which is better depends on the noise:

Spectrum in the Bell basis ($|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle$): recall each Bell state has definite $\langle XX \rangle, \langle ZZ \rangle, \langle YY \rangle = \pm 1$. We compute two valid witnesses for all basis states as follows:

Bell state	$\langle XX \rangle$	$\langle YY \rangle$	$\langle ZZ \rangle$	$\mathcal{W} = \frac{XX+ZZ}{2}$	$\mathcal{W}' = \frac{XX-YY}{2}$
Φ^+ (target)	+1	-1	+1	+1	+1
Φ^-	-1	+1	+1	0	-1
Ψ^+	+1	+1	-1	0	0
Ψ^-	-1	-1	-1	-1	0

So $\mathcal{W} = \text{diag}(1, 0, 0, -1)$ and $\mathcal{W}' = \frac{1}{2}(XX - YY) = \text{diag}(1, -1, 0, 0)$ — **both are** valid witnesses (correct spectral structure), both needing only **2 measurement settings** instead of the 3 required for exact $F = \frac{1}{4}(1 + \langle XX \rangle - \langle YY \rangle + \langle ZZ \rangle)$. Which is suitable for which target state can be observed from the below table:

Preparation	F	$\langle \mathcal{W} \rangle$	$\langle \mathcal{W}' \rangle$
$0.9 \Phi^+ + 0.1 \Psi^-$	0.9	0.8 (gap 0.1)	0.9 (gap 0, exact)
$0.9 \Phi^+ + 0.1 \Phi^-$	0.9	0.9 (gap 0, exact)	0.8 (gap 0.1)

Each witness has a blind spot (its 0-eigenvalue directions) and a sore spot (its -1 direction). $\mathcal{W}$ and $\mathcal{W}'$ are **incomparable** under $\preceq$ — neither dominates. Choosing a witness = choosing where to place the sore spot, ideally away from where the hardware's noise goes.

4. Where witnesses come from

4.1 The parent-Hamiltonian construction

Origin of the witness.

Let $\hat{H}$ have unique ground state $|\psi_t\rangle$ with energy E_0 , first excited energy E_1 , gap $\Delta = E_1 - E_0 > 0$. Then $\mathcal{W} = \frac{E_1 - \langle \hat{H} \rangle}{\Delta}$ is a fidelity witness for $|\psi_t\rangle$.

Proof. Expand ϱ_p in eigenstates of $\hat{H}$ with probabilities p_k ; note $p_0 = F$:

$$\langle \hat{H} \rangle = \sum_k E_k p_k \geq E_0 p_0 + E_1 \sum_{k \geq 1} p_k = E_0 F + E_1 (1 - F) \implies \boxed{F \geq \frac{E_1 - \langle \hat{H} \rangle}{\Delta}}$$

Condition (i): equality iff all population sits in the (unique) ground state. ■

Energy shift. Absolute energy is unobservable, so shift $\hat{H}' = \hat{H} - E_0 \mathbf{1}$ (then $E'_0 = 0, E'_1 = \Delta$):

$$\mathcal{W} = 1 - \frac{\langle \hat{H}' \rangle}{\Delta} \quad (E_0 = 0 \text{ convention})$$

The shift is free physically but **not free in bookkeeping**: applying $1 - \hat{H}/\Delta$ to an unshifted Hamiltonian with $E_0 < 0$ produces an *unsound* bound (it over-claims — e.g. it can output $F \geq 2$). So always shift first, or use the E_1 form.

Physical reading: *infidelity is bounded by excess energy in units of the gap*: $1 - F \leq (\langle \hat{H} \rangle - E_0)/\Delta$. Measuring energy is a proxy for measuring infidelity, at exchange rate Δ . Nearly-gapless parent Hamiltonians give weak witnesses (as bound degrades as $1/\Delta$).

Design checklist for a target $|\psi_t\rangle$:

- ✓ $\hat{H}|\psi_t\rangle = 0$ (target is ground state, $E_0 = 0$)
- ✓ ground state **unique** (else you certify a subspace, breaking (i))
- ✓ Δ **known or lower-bounded** — the hard part; for general local $\hat{H}$ this is QMA-hard. Underestimating Δ is safe (looser but sound); overestimating produces false certificates.
- ✓ few Pauli measurement settings

Example (stabilizer states). If $g_1, \dots, g_n$ are commuting Pauli generators with $g_i|\psi_t\rangle = |\psi_t\rangle$:

$$\hat{H} = \sum_{i=1}^n \frac{1 - g_i}{2}$$

Eigenvalues of $\hat{H}$ = number of violated stabilizers $\in \{0, 1, \dots, n\}$, so $E_0 = 0, \Delta = 1$ **exactly and for free** (a counting argument, no diagonalization), ground state unique. Note that stabilizers have eigenvalues ± 1 .

$$F \geq \frac{1}{2} \sum_{i=1}^n \langle g_i \rangle - \frac{n-2}{2}$$

Check on the Bell state $|\Phi^+\rangle$ (for this $g_1 = XX, g_2 = ZZ, \Delta = 1$): $\mathcal{W} = 1 - \langle \hat{H} \rangle = \frac{1}{2}(XX + ZZ)$ — exactly §3's witness, now *derived* instead of guessed. Its sore spot $|\Psi^-\rangle$ is the doubly-excited state ($E = 2$): the witness charges it linearly (-1) while fidelity charges it once.

4.2 The projector Proposition (general recipe)

Proposition. Let ϱ_t be any pure target, $0 < \Delta = \lambda_1 \leq \dots \leq \lambda_N$, and $P_1, \dots, P_N \succeq 0$ with $\varrho_t + \sum_{l=1}^N P_l = \mathbf{1}$, $\text{tr}[\varrho_t P_l] = 0 \forall l$.

Then $\mathcal{W} := 1 - \Delta^{-1} \sum_l \lambda_l P_l$ is a fidelity witness for ϱ_t , with $\Delta = \min_l \lambda_l$.

$P_l \succeq 0$ = each piece measures a nonnegative amount of error; $\text{tr}[\varrho_t P_l] = 0$ = none fires on the target (and by the PSD Lemma, $P_l|\psi_t\rangle = 0$); **completeness** $\varrho_t + \sum P_l = \mathbf{1}$ = every error direction is covered by something; $\lambda_l \geq \Delta$ = every error

is charged at least Δ .

Check of condition (i): $\langle \psi_t | \mathcal{W} | \psi_t \rangle = 1 - \Delta^{-1} \sum_l \lambda_l \underbrace{\langle \psi_t | P_l | \psi_t \rangle}_0 = 1 \checkmark$

Check of condition (ii): substitute completeness $1 = \varrho_t + \sum_l P_l$:

$$\varrho_t - \mathcal{W} = \varrho_t - 1 + \frac{1}{\Delta} \sum_l \lambda_l P_l = \sum_l \left(\frac{\lambda_l}{\Delta} - 1 \right) P_l \succeq 0 \quad \checkmark$$

(nonnegative coefficients $\times$ PSD; this is why $\Delta = \lambda_{\min}$ is the right normalization). And the gap becomes fully explicit:

$$F - \langle \mathcal{W} \rangle_{\varrho_p} = \sum_l \left(\frac{\lambda_l}{\Delta} - 1 \right) \text{tr}[P_l \varrho_p]$$

- the $\lambda_l = \Delta$ pieces contribute **zero** gap $\rightarrow$ blind spots (witness exact there);
- $\lambda_l \gg \Delta$ pieces are over-penalized by $\lambda_l/\Delta \rightarrow$ sore spots;
- all $\lambda_l = \Delta \Rightarrow \mathcal{W} = \varrho_t$ — perfectly tight, exponentially expensive. Every λ_l raised above Δ purchases measurement simplicity, paid in gap.

4.3 The equivalence

For $\hat{H} \succeq 0$ with $\hat{H}|\psi_t\rangle = 0$ and unique ground state, the following are **equivalent**:

1. $\exists \{P_l, \lambda_l\}$ as in the Proposition with $\hat{H} = \sum_l \lambda_l P_l$;
2. $\hat{H} \succeq \Delta(1 - \varrho_t)$;
3. spectral gap of $\hat{H} \geq \Delta$.

(1 $\Rightarrow$ 2): $\hat{H} - \Delta(1 - \varrho_t) = \sum_l (\lambda_l - \Delta) P_l \succeq 0$. (2 $\Rightarrow$ 3): sandwich with any $|\phi_\perp\rangle$. (3 $\Rightarrow$ 1): spectral decomposition on $\psi_t^\perp$. ■

So the certification task is exactly: certify $\hat{H} \succeq \Delta(1 - \varrho_t)$. The Proposition is not a search algorithm for $\{P_l, \lambda_l\}$. In practice we work backwards: pick a **measurable** $\hat{H}$ first (locality / budget decides this), then exhibit any decomposition to certify Δ — e.g. for stabilizers, label error sectors by syndrome and *count* ($\lambda_s = |s| \geq 1$). The λ_l are never chosen freely; they are the energies of the $\hat{H}$ you already picked, possibly coarse-grained downward (safe direction).

4.4 The transport lemma — witnesses travel with unitaries

✿ **Lemma.** If $\mathcal{W}_0$ is a fidelity witness for $|\psi_0\rangle$, then $\mathcal{W} = U\mathcal{W}_0U^\dagger$ is a fidelity witness for $U|\psi_0\rangle$.

Proof. (ii): conjugation preserves PSD — $\langle \phi | U M U^\dagger | \phi \rangle = \langle U^\dagger \phi | M | U^\dagger \phi \rangle \geq 0$ — so $\varrho_0 - \mathcal{W}_0 \succeq 0 \Rightarrow U\varrho_0U^\dagger - U\mathcal{W}_0U^\dagger \succeq 0$. (i): $\text{tr}[U\mathcal{W}_0U^\dagger \cdot U\varrho U^\dagger] = \text{tr}[\mathcal{W}_0\varrho]$ by cyclicity, and $\varrho \mapsto U\varrho U^\dagger$ is a bijection on states. Spectrum unchanged. ■

Why this is powerful: soundness (conditions i, ii) and the gap Δ are established **once, at a trivial reference state**, then transported to every $U|\psi_0\rangle$ for free. The open question becomes purely one of *measurability*: does $\mathcal{W} = U\mathcal{W}_0U^\dagger$ still have few Pauli terms? Generically no ($\sim 4^L$ terms). It stays cheap exactly when $\mathcal{W}_0$ and the generators of U live in a **polynomially-sized Lie algebra closed under commutation** — by the Hadamard lemma $UAU^\dagger = A + [G, A] + \frac{1}{2!}[G, [G, A]] + \dots$, conjugation can never leave the algebra. The fermionic Gaussian case (next) is the canonical instance: quadratic Majorana operators form $\mathfrak{so}(2L)$, dimension $\binom{2L}{2} = O(L^2)$, **at any circuit depth**.

5. The fermionic toolbox

5.1 Modes and the CAR algebra

L spinless fermionic modes (one orbital each, no spin label), annihilation/creation operators $f_j, f_j^\dagger$ obeying the **canonical anticommutation relations (CAR)**:

$$\{f_i, f_j^\dagger\} = \delta_{ij}, \quad \{f_i, f_j\} = \{f_i^\dagger, f_j^\dagger\} = 0$$

Setting $i = j$ in the second relation: $(f_j^\dagger)^2 = 0$ — **Pauli exclusion** built into the algebra; each mode holds 0 or 1 fermions. Note f_j is *not* Hermitian.

Number operator $n_j = f_j^\dagger f_j$: using the CAR, $n_j^2 = f_j^\dagger f_j f_j^\dagger f_j = f_j^\dagger (1 - f_j^\dagger f_j) f_j = n_j$ — idempotent Hermitian $\Rightarrow$ a **projector**, eigenvalues $\{0, 1\}$.

5.2 Majorana operators

$$m_{2j-1} = f_j + f_j^\dagger, \quad m_{2j} = -i(f_j - f_j^\dagger)$$

L modes $\Rightarrow 2L$ Majoranas (the fermionic analogue of splitting the SHO ladder operator into $\hat{x} \sim a + a^\dagger$ and $\hat{p} \sim -i(a - a^\dagger)$). Both are **Hermitian** — genuine observables. They generate a **Clifford algebra**:

$$\{m_j, m_k\} = 2\delta_{jk} \quad \Rightarrow \quad m_j^2 = 1, \text{ eigenvalues } \pm 1$$

The key single-mode identity:

$$i m_{2j-1} m_{2j} = 2n_j - 1$$

The prefactor i is essential: $m_{2j-1} m_{2j}$ alone is *anti*-Hermitian; the i makes it a legitimate ± 1 -valued observable.

Why quadratics are special (the Lie-algebra fact from §4.4): the set $\{i m_j m_k\}_{j < k}$ is closed under commutation and has $\binom{2L}{2} = L(2L-1) = O(L^2)$ elements — a Lie algebra isomorphic to $\mathfrak{so}(2L)$ (which has dimension $N(N-1)/2$ at $N = 2L$). *Polynomial*, not exponential.

5.3 Jordan–Wigner: fermions on qubits

At $L = 1$: $f = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ in basis $(|0\rangle, |1\rangle)$ gives $m_1 = X$, $m_2 = Y$, and $-im_1 m_2 = Z$. For $L > 1$, anticommutation across sites forces the **string**:

$$m_{2j-1} = Z_1 \cdots Z_{j-1} X_j, \quad m_{2j} = Z_1 \cdots Z_{j-1} Y_j, \quad f_j = \left(\prod_{k < j} Z_k \right) \frac{X_j + iY_j}{2}$$

Occupation dictionary:

$$n_j = \frac{1 - Z_j}{2} \quad \Rightarrow \quad Z_j |\omega\rangle = (-1)^{\omega_j} |\omega\rangle$$

- $\omega_j = 1 \iff$ mode j **occupied** (this is what $n_j |\omega\rangle = \omega_j |\omega\rangle$ means)
- $|\uparrow\rangle$ ($Z = +1$) = **empty** mode; the fermionic **vacuum** is $|\uparrow\rangle^{\otimes L}$, i.e. $\omega = (0, \dots, 0)$

5.4 Quadratic Hamiltonians and Gaussian unitaries

$$H = \frac{i}{4} \sum_{j,k=1}^{2L} A_{jk} m_j m_k, \quad A = -A^T \in \mathbb{R}^{2L \times 2L}$$

Any symmetric part of A dies against $m_j m_k$'s antisymmetry; the i ensures Hermiticity; the $1/4$ compensates the double count $j \leftrightarrow k$. "Free" Hamiltonian: no interaction terms like $f_j^\dagger f_k^\dagger f_k f_j = n_j n_k$.

$U = e^{-iHt}$ is a **fermionic Gaussian unitary**: it maps Gaussian states to Gaussian states.

U itself is a $2^L \times 2^L$ unitary — it is **not** in $SO(2L)$. What lives in $SO(2L)$ is its *action on the modes*, next.

Heisenberg evolution stays linear. Using $[m_a m_b, m_j] = 2(\delta_{bj} m_a - \delta_{aj} m_b)$ (pure Clifford algebra) and antisymmetry of A :

$$\frac{d}{dt} m_j(t) = i[H, m_j(t)] = \sum_a A_{ja} m_a(t) \implies \boxed{U^\dagger(t) m_j U(t) = \sum_k Q_{jk}(t) m_k, \quad Q(t) = e^{tA}}$$

A **closed linear ODE on only $2L$ operators**. Q is real orthogonal with $\det Q = e^{t \operatorname{tr} A} = 1$, so $Q \in SO(2L)$. *This is the entire reason Gaussian dynamics is classically simulable: $2L \times 2L$ matrix exponentials instead of $2^L \times 2^L$.*

5.5 The covariance matrix (fundamentals)

$$M_{jk}(\varrho) := \frac{i}{2} \operatorname{tr}([m_j, m_k] \varrho) \implies M_{jk} = i \langle m_j m_k \rangle \ (j \neq k), \quad M_{jj} = 0$$

M is **real, antisymmetric**, $2L \times 2L - O(L^2)$ numbers. Contents:

- diagonal blocks: occupations, $\langle n_j \rangle = \frac{1 + M_{2j-1, 2j}}{2}$
- off-diagonal blocks: hopping correlations $\langle f_j^\dagger f_k + \text{h.c.} \rangle$ and pairing correlations $\langle f_j^\dagger f_k^\dagger + \text{h.c.} \rangle$

Example ($L = 1$, vacuum $|0\rangle = |\uparrow\rangle$): $M_{12} = i \langle XY \rangle = i \langle iZ \rangle = -\langle Z \rangle = -1$, so $M = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\langle n \rangle = \frac{1+(-1)}{2} = 0$.
✓.

Three structural facts:

1. **Purity test:** $M^2 = -1 \iff \varrho$ is a *pure* Gaussian state $\iff MM^T = 1$ (orthogonal). $M^2 \neq -1 \implies$ mixed.

$$\sum_k M_{jk}^2 = (MM^T)_{jj} = 1 \xrightarrow{\text{C-S}} \sum_k |M_{jk}| \leq \sqrt{2L} \implies \boxed{|M(\varrho_i)| \leq \sqrt{2} L^{3/2}}$$

2. **Transformation law:** by bilinearity of M in the Majoranas and $U^\dagger m_j U = \sum_a Q_{ja} m_a$,

Q is *orthogonal*; M is antisymmetric.

$$M(U^\dagger \varrho U) = Q^T M(\varrho) Q \iff M(U \varrho U^\dagger) = Q M(\varrho) Q^T$$

3. **Fock states:** on $|\omega\rangle$, $M_{2k-1, 2k} = \langle 2n_k - 1 \rangle = 2\omega_k - 1$ and everything else vanishes:

$$\boxed{M_\omega = \bigoplus_{k=1}^L (1 - 2\omega_k) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}$$

5.6 Wick's theorem

For Gaussian states, all higher correlations factor into pairs (with fermionic signs):

$$\langle m_1 m_2 m_3 m_4 \rangle = \langle m_1 m_2 \rangle \langle m_3 m_4 \rangle - \langle m_1 m_3 \rangle \langle m_2 m_4 \rangle + \langle m_1 m_4 \rangle \langle m_2 m_3 \rangle$$

and in general $\langle m_{j_1} \cdots m_{j_{2p}} \rangle = \operatorname{Pf}(C)$, the **Pfaffian** of the antisymmetric matrix of pair correlators ($\operatorname{Pf}(C)^2 = \det C$). So a Gaussian state is characterized by $O(L^2)$ two-point functions instead of 2^L amplitudes — the same way a mean-zero classical Gaussian $\mathcal{N}(0, \sigma^2)$ is fixed by $\langle x^2 \rangle = \sigma^2$ (then $\langle x^4 \rangle = 3\sigma^4, \dots$). Hence the name.

6. The witness for fermionic Gaussian targets

6.1 The error-counting Hamiltonian $n^{(\omega)}$

Target class: $|\psi_t\rangle = U|\omega\rangle$ with U Gaussian and $|\omega\rangle = |\omega_1 \cdots \omega_L\rangle$ a Fock state ($\omega_j = 1 \iff$ occupied). By the transport lemma we only need a witness at $|\omega\rangle$, i.e. a parent Hamiltonian with ground state $|\omega\rangle$, unique, known gap. Build it mode by mode — we want an operator that charges 0 if mode j agrees with ω_j and 1 if it disagrees:

$$\text{mode } j : \quad (1 - \omega_j) n_j + \omega_j (1 - n_j) = \begin{cases} n_j & \omega_j = 0 \text{ (penalize occupation)} \\ 1 - n_j & \omega_j = 1 \text{ (penalize vacancy)} \end{cases}$$

$$n^{(\omega)} = \sum_{j=1}^L \left[(1 - \omega_j) n_j + \omega_j (1 - n_j) \right] \quad \text{number of incorrectly occupied modes}$$

Under JW ($n_j = (1 - Z_j)/2$) this is a fully diagonal operator:

$$n^{(\omega)} = \sum_{j=1}^L \frac{1 - (-1)^{\omega_j} Z_j}{2} \implies \boxed{n^{(\omega)} |\nu\rangle = d(\omega, \nu) |\nu\rangle}$$

with $d = \text{Hamming distance}$. Checklist of §4.1:

- ground state $|\omega\rangle$, energy 0, **unique** ✓ (only $\nu = \omega$ has $d = 0$)
- spectrum $\{0, 1, \dots, L\} \implies$ gap $\Delta = 1$, **exact and free** ✓ (a counting argument, like stabilizers)
- worked example $L = 4$, Target $\omega = (0101)$: $n^{(\omega)} = n_1 + (1 - n_2) + n_3 + (1 - n_4)$, spectrum $\{0, 1, 2, 3, 4\}$ ✓

💡 Every one of the 2^L Fock states is itself a Gaussian state, so this construction covers the whole reference family at once; U then transports it anywhere in the Gaussian manifold.

6.2 The witness for $|\psi_t\rangle$

$$\boxed{\mathcal{W} = U(1 - n^{(\omega)})U^\dagger} \quad (\text{parent Hamiltonian } n^{(\omega)}, \Delta = 1, \text{ transported by } U)$$

Its expectation has a beautiful direct meaning:

$$F_{\mathcal{W}}(\varrho_p) := \text{tr}[\mathcal{W}\varrho_p] = 1 - \text{tr}[Un^{(\omega)}U^\dagger\varrho_p] = \boxed{1 - n_{\perp}(\varrho_p)}$$

where $n_{\perp}(\varrho_p) = \text{mean number of misplaced fermions}$, measured in the rotated frame. *Fidelity is lower-bounded by one minus the average count of wrongly-filled modes.*

6.3 From operator to covariance matrices

This is the computational heart of the Gluza et al. paper: turning $\text{tr}[\mathcal{W}\varrho_p]$ into pure $2L \times 2L$ linear algebra.

Step 1 — write $n^{(\omega)}$ in Majoranas. From $im_{2j-1}m_{2j} = 2n_j - 1$, i.e. $n_j = \frac{i}{4}[m_{2j-1}, m_{2j}] + \frac{1}{2}$:

$$n^{(\omega)} = \sum_{j=1}^L \left[\frac{1}{2} + \frac{i}{4}(1 - 2\omega_j) [m_{2j-1}, m_{2j}] \right]$$

Step 2 — take the expectation, absorbing U into the state. Using cyclicity, $\text{tr}[Un^{(\omega)}U^\dagger\varrho_p] = \text{tr}[n^{(\omega)}U^\dagger\varrho_pU]$. Writing $\tilde{M} := M(U^\dagger\varrho_pU)$ and recalling $\text{tr}([m_j, m_k]\varrho) = -2i M_{jk}(\varrho)$:

$$F_{\mathcal{W}} = 1 - \frac{L}{2} - \frac{1}{2} \sum_{k=1}^L (1 - 2\omega_k) \tilde{M}_{2k-1,2k}$$

Step 3 — package the sum as a trace against M_ω . A two-line block computation gives $\sum_k (1 - 2\omega_k) \tilde{M}_{2k-1,2k} = \frac{1}{2} \text{tr}[\tilde{M} M_\omega]$, hence

$$F_{\mathcal{W}} = 1 - \frac{L}{2} - \frac{1}{4} \text{tr}[\tilde{M} M_\omega]$$

Step 4 — move the frame rotation onto the target. $\tilde{M} = Q^T M(\varrho_p) Q$ and $Q M_\omega Q^T = M(U|\omega\rangle\langle\omega|U^\dagger) = M(\varrho_t)$, so by cyclicity:

$$F_{\mathcal{W}}(\varrho_p) = 1 - \frac{L}{2} - \frac{1}{4} \text{tr}[M(\varrho_p) M(\varrho_t)] \rightarrow \quad (1)$$

Step 5 — the paper's form. Using $M_t^T = -M_t$ and purity $M_t^2 = -1 \Rightarrow \text{tr}[M_t^T M_t] = 2L$:

$$F_{\mathcal{W}}(\varrho_p) = 1 + \frac{1}{4} \text{tr}[(M(\varrho_p) - M(\varrho_t))^T M(\varrho_t)]$$

Both forms verified equal, and $F_{\mathcal{W}}(\varrho_t) = 1$, at machine precision.

💡 Three remarks worth internalizing.

1. **No Wick's theorem was used.** $n^{(\omega)}$ is quadratic, so only two-point functions of ϱ_p ever appear — by *linearity*, not Gaussianity. Hence ϱ_p may be **arbitrary**: mixed, non-Gaussian, entangled with a bath. Only the *target* must be Gaussian (purity of $\tilde{M}_t$ entered once, in Step 5).
2. **Where Gaussianity of U is truly needed:** to make the frame rotation *classical* ($Q \in SO(2L)$ acting on matrices). For non-Gaussian U we'd have to physically undo U on the device before measuring — a compute-uncompute scheme that certifies the wrong thing (errors of $U^\dagger U$ can cancel).
3. The witness costs **at most $2L^2 - L$ numbers** of $M(\varrho_p)$ — versus $\sim 4^L$ Pauli coefficients for direct fidelity estimation of a generic Gaussian state. This is the exponential → polynomial collapse.

6.4 The systematic gap

Decompose any preparation along the target (this uses **only linearity** — no positivity):

$$\varrho_p = F \varrho_t + (1 - F) \varrho_\perp, \quad \varrho_\perp := \frac{\varrho_p - F \varrho_t}{1 - F}$$

$\varrho_\perp$ is trace-one, Hermitian, and Hilbert-Schmidt-orthogonal to the target ($\text{tr}[\varrho_t \varrho_\perp] = 0$ orthogonality works only if target is pure), but it is **not necessarily a valid density matrix** — coherences between $|\psi_t\rangle$ and its complement make it non-PSD.

Why the above decomposition is valid?

Hermitian operators are equipped with with the Hilbert-Schmidt inner product $\langle A, B \rangle = \text{tr}[A^\dagger B]$. Then:

- a **pure** state is a *unit vector*: $\langle \varrho_t, \varrho_t \rangle = 1$
- fidelity is literally an **inner product**: $F = \langle \varrho_t, \varrho_p \rangle$
- therefore $F \varrho_t$ is exactly the **orthogonal projection** of ϱ_p onto the target's ray

$$\varrho_p = \underbrace{\langle \varrho_t, \varrho_p \rangle \varrho_t}_{\text{parallel}} + \underbrace{(\varrho_p - F \varrho_t)}_{\text{perpendicular}}$$

This is ordinary vector decomposition. The $(1 - F)$ is just a normalization pulled out so the residual has trace 1.

Since $n_{\perp}(\varrho_t) = 0$ (the target has no misplaced fermions) and $n_{\perp}$ is linear:

$$n_{\perp}(\varrho_p) = (1 - F) n_{\perp}(\varrho_{\perp}) \quad \implies \quad \boxed{F - F_{\mathcal{W}}(\varrho_p) = (1 - F) (n_{\perp}(\varrho_{\perp}) - 1)}$$

$n_{\perp}(\varrho_{\perp}) = \text{average excitations per unit of error.}$

Error type	$n_{\perp}(\varrho_{\perp})$	Gap $F - F_{\mathcal{W}}$	Verdict
single excitations (mode loss, T_1)	1	0	witness exact
pair creation (coherent quadratic error, Trotter)	2	$1 - F$	$F_{\mathcal{W}} \approx 2F - 1$
all L modes wrong (symmetry-broken partner)	L	$(L - 1)(1 - F)$	catastrophic

This one formula explains all witness pessimism seen in practice.

7. What to measure for computing witness, concretely

7.1 The support set Ω

$$\Omega := \{(j, k) : M_{jk}(\varrho_t) \neq 0, 1 \leq j < k \leq 2L\}, \quad |\Omega| \leq 2L^2 - L$$

$M(\varrho_t)$ is computed **classically** ($O(L^3)$): build M_{ω} , exponentiate A , conjugate). Entries of $M(\varrho_p)$ where M_t vanishes are multiplied by zero in Eq. (1) — **the target dictates the measurement schedule**;

7.2 Jordan–Wigner dictionary for the correlators

Each $i m_j m_k$ ($j < k$) maps to a single Pauli string: endpoint operators from the parities of j, k , a Z -string strictly between the *sites* (from incomplete cancellation of the two JW strings), phases $\mp i$ absorbed:

parity of j	parity of k	$m_j m_k$ (sites $\lceil j/2 \rceil < \lceil k/2 \rceil$)
odd	odd	$-i \sigma^y (Z \cdots Z) \sigma^x$
odd	even	$-i \sigma^y (Z \cdots Z) \sigma^y$
even	odd	$+i \sigma^x (Z \cdots Z) \sigma^x$
even	even	$+i \sigma^x (Z \cdots Z) \sigma^y$

(Derivable in two lines from $m_{2k-1} = (\prod_{k' < k} Z_{k'}) X_k$, $m_{2k} = (\prod_{k' < k} Z_{k'}) Y_k$ and $\sigma^a \sigma^b = \delta_{ab} 1 + i \epsilon_{abc} \sigma^c$.) So $i m_j m_k$ is Hermitian with eigenvalues $\beta = \pm 1$, and its projectors $\hat{m}_{j,k}^{(\beta)} = \frac{1}{2}(1 + \beta i m_j m_k)$ each have rank 2^{L-1} .

7.3 Extracting β in hardware — depth-1

1. $i m_j m_k \xrightarrow{\text{JW}}$ Pauli string (table above).
2. **Basis change, single-qubit only:** H on X endpoints, $HS^{\dagger}$ on Y endpoints ($X = HZH^{\dagger}$, $Y = (HS^{\dagger})Z(HS^{\dagger})^{\dagger}$); nothing on the Z string.
3. Measure **all** qubits in the Z basis $\rightarrow$ bitstring $b \in \{0, 1\}^L$.
4. $\beta = (-1)^{b_j \oplus b_{j+1} \oplus \cdots \oplus b_k}$ — parity over the support *only*; qubits outside $[j, k]$ are ignored.

One circuit layer of depth 1, per shot. Born rule: $\Pr(\beta | j, k) = \text{tr}[\hat{m}_{j,k}^{(\beta)} \varrho_p]$, so $\mathbb{E}[\beta | j, k] = \langle i m_j m_k \rangle_{\varrho_p} = M_{jk}(\varrho_p)$.

8. Importance sampling

8.1 Why sample at all

Estimating all $|\Omega| \sim 2L^2$ correlators separately, each to precision good enough that the *sum* has error ϵ , makes the shot count scale with $|\Omega|$ (variances add). Better: build **one random variable whose mean is the whole sum**, and let the sampling process itself perform the weighted summation.

8.2 The estimator

Target-weighted distribution over correlators:

$$P_{j,k} = \frac{|M_{jk}(\varrho_t)|}{|M(\varrho_t)|}, \quad |M(\varrho_t)| := \sum_{(j,k) \in \Omega} |M_{jk}(\varrho_t)| \leq 2L^2$$

Per shot: draw $(j, k) \sim P$ (classical, redrawn *every* shot) $\rightarrow$ prepare $\varrho_p \rightarrow$ measure $\beta = \pm 1$ as in §7.3 $\rightarrow$ record

$$X_{\beta,j,k} := 2 |M(\varrho_t)| \beta \operatorname{sgn}[M_{jk}(\varrho_t)]$$

Unbiasedness (the whole point):

$$\mathbb{E}[X] = \sum_{(j,k) \in \Omega} P_{j,k} \sum_{\beta=\pm 1} \Pr(\beta|j,k) X_{\beta,j,k} = 2 \sum_{(j,k) \in \Omega} M_{jk}(\varrho_p) M_{jk}(\varrho_t) = \operatorname{tr}[M(\varrho_p)^T M(\varrho_t)]$$

(the $|M_t|$'s cancel between $P_{j,k}$ and X ; $|M_{jk}(\varrho_t)| \cdot \operatorname{sgn} = M_{jk}(\varrho_t)$; $\mathbb{E}[\beta|j,k] = M_{jk}(\varrho_p)$; the factor 2 converts the upper-triangle sum into the full antisymmetric trace, and extending beyond Ω is free since $M_t = 0$ there). Then, from Eq. (1):

$$F_{\mathcal{W}}^* = 1 + \frac{1}{4}(\bar{X} - 2L), \quad \bar{X} = \frac{1}{N} \sum_{s=1}^N X^{(s)} \quad (2)$$

Sanity check: perfect preparation gives $\bar{X} \rightarrow \operatorname{tr}[M_t^T M_t] = 2L$, so $F_{\mathcal{W}}^* \rightarrow 1$. ✓

💡 **This is Monte Carlo, not tomography.** One correlator per shot, redrawn each shot. **No individual $M_{jk}(\varrho_p)$ is ever estimated well**; only the weighted sum converges. Entries where the target is large get visited often (hence *importance* sampling); this is exactly the $N_P \propto |c_P|$ optimum of §1.3, transplanted from the (4^L -dimensional) Pauli basis to the ($2L^2$ -dimensional) correlator basis. That change of basis is the entire exponential saving.

8.3 Sample complexity

Theorem

$\epsilon > 0$

$\delta \in (0, 1)$ the estimate satisfies

$\Pr(|F_{\mathcal{W}} - F_{\mathcal{W}}^*| \leq \epsilon) \geq 1 - \delta$ with

$$N_{\epsilon, \delta} \leq \left\lceil \frac{\ln(2/\delta) |M(\varrho_t)|^2}{2\epsilon^2} \right\rceil$$

Proof. X takes values in $\{\pm 2|M_t|\}$, so Hoeffding applies with range $R = 4|M_t|$:

$$\Pr(|\bar{X} - \mathbb{E}X| \geq t) \leq 2 \exp\left(\frac{-2Nt^2}{R^2}\right)$$

From Eq (1) and Eq (2) $|F_{\mathcal{W}} - F_{\mathcal{W}}^*| = \frac{1}{4}|\bar{X} - \mathbb{E}X|$, precision ϵ on $F_{\mathcal{W}}$ means $t = 4\epsilon$. The 16's cancel:

$$\delta = 2 \exp\left(\frac{-2N(4\epsilon)^2}{(4|M_t|)^2}\right) = 2 \exp\left(\frac{-2N\epsilon^2}{|M_t|^2}\right) \implies N = \frac{\ln(2/\delta) |M_t|^2}{2\epsilon^2} \quad (\text{ceil}). \blacksquare$$

Cost is governed entirely by the classical, pre-computable number $|M(\varrho_t)|$:

Regime	$ M_t $	N
worst case	$\sqrt{2}L^{(3/2)}$	$O(L^3/\epsilon^2)$
critical quench, $t = L/8$	$\approx 2.35 L^{1.40}$	$O(L^{2.8}/\epsilon^2)$
gapped local ground state (clustering)	$\sim L \log L$	$\tilde{O}(L^2/\epsilon^2)$

You know the full shot budget before touching the device.

9. Robust certification

9.1 The decision rule

Fix a threshold fidelity F_T and run the estimator with precision ϵ , confidence $1 - \delta$:

$$\text{Accept} \iff F_{\mathcal{W}}^* \geq F_T + \epsilon$$

Design constraints: $0 < \epsilon < \frac{1-F_T}{2}$, $2\epsilon < \Delta_c < 1 - F_T$, $\delta > 0 = \text{max failure probability}$. (Δ_c = certification fidelity gap: the deliberately-tolerated grey zone as this gap is systematic.)

9.2 Rejection is unconditional — the witness never falsely certifies

Suppose $F < F_T$. With probability $\geq 1 - \delta$, $F_{\mathcal{W}}^* - \epsilon \leq F_{\mathcal{W}}$, and $F_{\mathcal{W}} \leq F$ always. Chain:

$$F_{\mathcal{W}}^* - \epsilon \leq F_{\mathcal{W}} \leq F < F_T \implies F_{\mathcal{W}}^* < F_T + \epsilon \implies \text{Reject}$$

No assumption on the noise anywhere. A bad device cannot pass, period. Equivalently, on acceptance you may claim, with confidence $1 - \delta$: $F \geq F_{\mathcal{W}}^* - \epsilon$.

9.3 Acceptance needs an assumption — the class $\mathcal{S}_{\perp}$

Because of the systematic gap (§6.4), a *good* state can still be rejected if its noise excites many modes. Guaranteed acceptance for $F \geq F_T + \Delta_c$ holds only within the class

$$\mathcal{S}_{\perp}(\Delta_c, \epsilon) = \left\{ \varrho_p : n_{\perp}(\varrho_{\perp}) \leq n_{\perp,T} \right\}, \quad n_{\perp,T}(\Delta_c, \epsilon) = \frac{1 - F_T - 2\epsilon}{1 - F_T - \Delta_c}$$

(> 1 automatically since $\Delta_c > 2\epsilon$.)

Proof of acceptance: if $F \geq F_T + \Delta_c$ and $n_{\perp}(\varrho_{\perp}) \leq n_{\perp,T}$,

$$F_{\mathcal{W}} = 1 - (1 - F)n_{\perp}(\varrho_{\perp}) \geq 1 - (1 - F_T - \Delta_c)n_{\perp,T} = F_T + 2\epsilon$$

and $F_{\mathcal{W}}^* \geq F_{\mathcal{W}} - \epsilon \geq F_T + \epsilon \implies \text{accept}$. ■ ($n_{\perp,T}$ is *reverse-engineered* to make exactly this cancellation happen.)

Summary of the guarantee:

True fidelity	Verdict	Assumption needed
$F < F_T$	always rejects	none
$F \geq F_T + \Delta_c$	accepts	$\varrho_p \in \mathcal{S}_{\perp}$
$F \in [F_T, F_T + \Delta_c)$	either (grey area — “don’t care”)	—

9.4 Capacity vs demand — using it in practice

$n_{\perp,T}$ is the **test's capacity** (computed from chosen F_T, Δ_c, ϵ). $n_{\perp}(\varrho_{\perp})$ is the **device's demand** (a physical fact). The theorem applies iff demand $\leq$ capacity. Two ways to run it:

- **Forward:** pick $\Delta_c \rightarrow$ read off which noise characters are covered.
- **Backward (experimentalist's mode):** assume a bound $n_{\perp}^{\max}$ on your noise character $\rightarrow$ solve $n_{\perp,T} \geq n_{\perp}^{\max}$ for the minimum grey zone:

$$\Delta_c \geq (1 - F_T) \frac{n - 1}{n} + \frac{2\epsilon}{n}, \quad n = n_{\perp}^{\max}$$

The demand side is provably not measurable from witness data: $n_{\perp}(\varrho_{\perp}) = (1 - F_W)/(1 - F)$ — numerator free ($= 1 - F_W^*$), but the denominator needs an *upper* bound on F , and the witness only ever gives lower bounds; substituting $F \geq F_W$ yields only the trivial $n_{\perp}(\varrho_{\perp}) \geq 1$. So $n_{\perp}^{\max}$ is a **declared assumption**, justified by: (1) classical simulation of the error model (Trotter: ≈ 2 , see Appendix B); (2) structure of the error generator (single-mode incoherent $\rightarrow 1$; quadratic coherent $\rightarrow$ pair creation $\rightarrow 2$; symmetry-breaking $\rightarrow L$); (3) small- L calibration + verified L -independence; (4) worst-case fallback $n_{\perp}^{\max} = L$, which is sound but forces $\Delta_c \approx 1 - F_T$ (nearly vacuum — guaranteed acceptance then demands $1 - F \lesssim \frac{(1-F_T)-2\epsilon}{L}$).

9.5 Non-triviality: when is the bound even informative?

$F_W = 1 - n_{\perp}$ can be negative (vacuous). Guarantee it isn't:

🔴 **Note.** If $F > 1 - L^{-2}$ then $F_W(\varrho_p) \geq 0$.

Proof. Work in the rotated frame $\tilde{\varrho} = U^\dagger \varrho U$ (trace distance and fidelity are unitarily invariant). $n^{(\omega)} = \sum_{k=1}^L P_k$ with each P_k ($0 \preceq P_k \preceq 1$) a single-mode error projector, and $\text{tr}[P_k \tilde{\varrho}_t] = 0$. The operational meaning of trace distance — $|\text{tr}[P(\varrho - \sigma)]| \leq D(\varrho, \sigma)$ for any $0 \preceq P \preceq 1$ — applied to each term, then Fuchs–van de Graaf $D \leq \sqrt{1 - F}$:

$$n_{\perp}(\varrho_p) = \sum_{k=1}^L \text{tr}[P_k(\tilde{\varrho}_p - \tilde{\varrho}_t)] \leq L \cdot D(\varrho_p, \varrho_t) \leq L\sqrt{1 - F} < L \cdot L^{-1} = 1 \implies F_W > 0 \blacksquare$$

(The bound $n_{\perp} \leq L\sqrt{1 - F}$ was verified numerically on test states. The cruder Hölder route $|\text{tr}[A(\varrho - \sigma)]| \leq \|A\|_{\infty} \|\varrho - \sigma\|_1$ with $\|n^{(\omega)}\|_{\infty} = L$ gives a factor-2-worse constant — the per-projector step matters.) In practice the worst case is very pessimistic: for realistic noise with $n_{\perp}(\varrho_{\perp}) \approx 2$, the bound is informative already for $F \gtrsim \frac{1}{2}$, independent of L .

10. Application: quenched transverse-field Ising / XY chain

10.1 The model and its coupling matrix

$$H_{\text{spin}} = - \sum_{k=1}^{L-1} (J_k^x \sigma_k^x \sigma_{k+1}^x + J_k^y \sigma_k^y \sigma_{k+1}^y) - \sum_{k=1}^L B_k \sigma_k^z \quad (\text{open boundary})$$

JW maps each term to a Majorana quadratic — the strings cancel between neighbours:

$$\sigma_k^z = -i m_{2k-1} m_{2k}, \quad \sigma_k^x \sigma_{k+1}^x = -i m_{2k} m_{2k+1}, \quad \sigma_k^y \sigma_{k+1}^y = +i m_{2k-1} m_{2k+2}$$

A term $i c m_a m_b$ ($a < b$) in H corresponds to $A_{ab} = 2c$, $A_{ba} = -2c$. Hence the three families of entries:

$$A_{2k-1,2k} = 2B_k, \quad A_{2k,2k+1} = 2J_k^x, \quad A_{2k-1,2k+2} = -2J_k^y$$

Ising case ($J^y = 0$): the dimerized Majorana chain. A is tridiagonal — $2L$ Majoranas on a line with alternating bonds:

$$m_1 \overset{2B}{-} m_2 \overset{2J}{-} m_3 \overset{2B}{-} m_4 \overset{2J}{-} m_5 \overset{2B}{-} m_6 \cdots$$

(B -bonds *within* a site, J -bonds *between* sites $\rightarrow$ the Kitaev-chain picture.) At $J = 0$: L disjoint dimers, ground state the product $|\uparrow\rangle^{\otimes L}$ (= fermionic vacuum). The competition B vs J — pair Majoranas on-site or across sites — is the phase transition.²

[From the Quadratic Hamiltonian to the Dispersion Relation](#)

10.2 Spectrum of A is the physics

A real antisymmetric $\Rightarrow$ eigenvalues $\pm i\epsilon_k$, $\epsilon_k \geq 0$ (Appendix A: canonical form). In the normal-mode basis:

$$H = \sum_{k=1}^L \epsilon_k \left(\tilde{n}_k - \frac{1}{2} \right), \quad E(\tilde{n}) = \sum_k \epsilon_k \left(\tilde{n}_k - \frac{1}{2} \right), \quad \tilde{n}_k \in \{0, 1\}$$

Ground state: all $\tilde{n}_k = 0$, $E_0 = -\frac{1}{2} \sum_k \epsilon_k$. First excitation: flip the cheapest mode $\Rightarrow$ **physical gap** $\Delta_{\text{spin}} = \min_k \epsilon_k$.

(Sanity check, 1 site, $J = 0$: $A = \begin{pmatrix} 0 & 2B \\ -2B & 0 \end{pmatrix}$, $\epsilon = 2B$; directly $H = -B\sigma^z = 2B(n - \frac{1}{2})$ ✓.)

Uniform chain, bulk dispersion:

$$\boxed{\epsilon(q) = 2\sqrt{J^2 + B^2 - 2JB \cos q}} \quad \Delta_{\text{spin}} = 2|J - B|$$

Circle picture: $\epsilon(q) = |z(q)|$ with $z(q) = 2B - 2Je^{-iq}$ — a circle of radius $2J$ centered at $2B$. $B > J$: circle excludes the origin (trivial/paramagnetic, gapped). $B < J$: circle encloses the origin (winding number 1 — topological/ferromagnetic; with open boundaries, two edge Majoranas whose splitting $\sim e^{-L/\xi}$, $\xi = 1/\ln(J/B)$, is the true lowest ϵ). $B = J$: circle through the origin — **critical**, gap closes as $q \rightarrow 0$, linearly: $\epsilon(q) \approx v|q|$ with velocity $v = 2J$. On a finite open chain at criticality the gap closes only as $\Delta_{\text{spin}} \sim \pi v/(2L)$.

10.3 The paper's benchmark: sudden quench at criticality

Protocol (Fig. 2 of the paper): prepare $|\uparrow\rangle^{\otimes L}$ (the B -only ground state = vacuum, trivially Gaussian) and evolve with the **critical** Hamiltonian $J = B = 1$ for time $t = L/8$:

$$|\psi_t\rangle = e^{-iH(J=B)t} |\uparrow\rangle^{\otimes L} \quad \Rightarrow \quad M(\varrho_t) = Q(t) M_\uparrow Q(t)^T, \quad Q(t) = e^{tA(J,B)}$$

Why this is the *adversarial* test case, on purpose:

- **Sudden quench** = the initial state is an eigenstate of neither part; energy $\sim L$ above the ground state is injected, and every mode is populated. Nothing relaxes (unitary evolution conserves mode populations).
- **Criticality** = gapless spectrum, algebraic correlations, no clustering argument available — the regime where $|M_t|$ is *largest*.
- **Light cone**: quasiparticle pairs emitted everywhere at $t = 0$ propagate with group velocity $|v_g(q)| = |\partial_q \epsilon| \leq v = 2J$; correlations M_{jk} switch on when the *pair* separation reaches the distance, i.e. front at site-distance $d \approx 2vt = 4Jt$. Choosing $t = L/8$ puts the front at $d = L/2$ — correlations across half the chain: entangled enough to be hard, structured enough to see the cone. (Wavefront position confirmed numerically; at $L = 60$ the light-cone structure needs Trotter number $T \gtrsim 40$ to resolve.)

10.4 Hardware side: Trotterized preparation

The device implements first-order Trotter — in mode space, exactly:

$$M(\varrho_p) = Q_T M_\uparrow Q_T^T, \quad Q_T = \left[e^{\Delta t A(J)} e^{\Delta t A(B)} \right]^T, \quad \Delta t = t/T$$

Both exponentials are cheap and *exact* individually ($A(B)$ = disjoint 2×2 blocks; $A(J)$ = disjoint dimers on the shifted pairing — closed-form rotations). So Trotter error is a **coherent Gaussian** error: ϱ_p is still a pure Gaussian state, just the wrong one — the ideal calibration setting, since F , F_W , and $n_\perp(\varrho_\perp)$ are all classically computable and comparable (Appendix).

Exact Gaussian overlap

For two pure Gaussian states,

$$F = \left| \det \frac{M_t + M_p}{2} \right|^{1/2}$$

Appendix. Trotter error is a 2-excitation noise (calibration data)

Exact classical computation on the critical quench (T scaled to hold F roughly fixed):

L	6	10	16	24	40	60	100
$n_\perp(\varrho_\perp)$	2.026	2.063	2.057	2.062	2.052	2.049	2.046

Flat to $\sim 1\%$ over a $17\times$ range of L — a small- L calibration ($L = 6$, trivially exact-diagonalizable) predicts $L = 100$. **The L -independence, not the value, is what you must check** before extrapolating an $n_\perp^{\max}$ assumption. Analytic cross-check for a different noise (independent per-mode flips with probability p): $n_\perp(\varrho_\perp) = \frac{Lp}{1 - (1-p)^L} \rightarrow 1$ for sparse errors ($Lp \ll 1$) — witness essentially exact exactly in the regime where certification is interesting.